

Optimal trend inflation in an open economy[☆]Daeha Cho^a, Jung Hyun Kim^b, Kwang Hwan Kim^{b,*}, Suk Joon Kim^c^a College of Economics and Finance, Hanyang University, South Korea^b School of Economics, Yonsei University, South Korea^c Department of Economics, New York University, United States of America

ARTICLE INFO

JEL classification:

E32
E58
F41
F45

Keywords:

Zero lower bound
Welfare
Optimal inflation target
Flexible exchange rate
Monetary union

ABSTRACT

We study the optimal inflation target in an open economy with a zero lower bound (ZLB) on nominal interest rates calibrated to the Euro area. When uncovered interest rate parity (UIP) holds, the optimal inflation target is smaller than in a closed economy. The key to this result is that real interest rates at the ZLB increase less as trade openness increases. This less pronounced increase in real interest rates in an open economy mitigates the contraction in aggregate demand, thus reducing the cost of ZLB. Additionally, the optimal inflation target in a monetary union is lower than in a flexible exchange rate regime: forming a monetary union results in a decrease in the optimal inflation rate by 0.24%, thereby increasing per-period welfare by 0.07%. When departures from UIP are significant, the optimal inflation target may be higher than in a closed economy, and the desirability of a monetary union increases further.

1. Introduction

The COVID-19 pandemic has once again pushed several nations into the zero lower bound (ZLB), leaving central banks with limited policy tools for stimulating economic activity. Some economists have argued that central banks should aim for higher target inflation rates than those deemed desirable under normal circumstances in order to allow more room to ease monetary policy during times of crisis (Blanchard, 2022; L'Huillier and Schoenle, 2023). Because raising the inflation target is costly in sticky-price models, the optimal inflation target is set to balance the cost and benefit of doing so, with the benefit depending on the cost of ZLB.

However, most existing studies investigating the optimal inflation target overlook international factors such as trade openness and relative prices.¹ As discussed in the international macroeconomic literature, these factors play an important role in changing the propagation of shocks studied in closed economy settings. Given that the world is integrated with respect to goods markets to some extent, a crucial question is whether international factors significantly alter the cost of hitting the ZLB and, thus, the optimal inflation target.

In this paper, we demonstrate that trade openness and the exchange rate regime – specifically, a flexible exchange rate regime and a monetary union – have significant impacts on the optimal inflation target. To this end, we build a two-country New Keynesian model with positive steady-state inflation (or trend inflation) calibrated to the Euro area. We use this model to investigate how trade openness, parameterized by the degree of home bias in the consumption basket, affects the optimal inflation target in each exchange

[☆] We thank the editor Yuriy Gorodnichenko, associate editor Johannes Wieland, and an anonymous referee for their insightful comments and suggestions, which have substantially improved the paper. We also appreciate the participants of ERMAS 2023, KES Macro Conference, and KES Summer Meetings for their comments. We acknowledge financial support from the R.K. Cho Research Cluster Program and the research fund of Hanyang University (HY-202300000002640).

* Correspondence to: School of Economics, Yonsei University, 50 Yonsei-Ro, Seodaemun-Gu, Seoul, South Korea.

E-mail addresses: daehac@hanyang.ac.kr (D. Cho), coby_kim27@yonsei.ac.kr (J.H. Kim), kimkh01@yonsei.ac.kr (K.H. Kim), sk10040@nyu.edu (S.J. Kim).

¹ Earlier contributions that investigate the optimal inflation target are Kim and Ruge-Murcia (2009), Adam and Weber (2019), Coibion et al. (2012), and Blanco (2021), among many others.

rate regime. Finally, we compare the welfare of two exchange rate regimes, each with its optimal inflation target, to evaluate the desirability of forming a monetary union in the Euro area.

The first results of this paper are derived by assuming that the uncovered interest rate parity (UIP) holds. The optimal inflation target is smaller in an open economy than in a closed economy—2.09% in an open economy and 2.67% in a closed economy. In addition, the optimal inflation target under a flexible exchange rate regime is higher than that under a monetary union. Specifically, if the core and periphery in the Euro area form a monetary union, the optimal inflation rate is reduced by 0.24%, thereby increasing per-period welfare by 0.07%.

The lower optimal inflation rate in an open economy arises because aggregate demand contraction at the ZLB is less severe in an open economy than in a closed economy. In a closed economy, excess savings triggered by a sizable negative Home demand shock lead to a fall in expected inflation at the liquidity trap, causing Home real interest rates to rise and Home consumption to contract. However, in an open economy, since part of the excess savings flows to Foreign in the form of Home net exports, output in Home contracts less than it would in a closed economy. As the deflationary expectations are less severe, Home real interest rates increase less, leading to less contraction in consumption. Additionally, in accordance with the UIP condition, a smaller increase in relative real rates mitigates the perverse movement of the terms of trade, thereby facilitating an increase in Home net exports. As a result, the benefits of increasing trend inflation and, consequently, the optimal inflation target in an open economy are smaller than those in a closed economy.

The higher optimal inflation target in a flexible exchange rate regime than in a monetary union originates from the fact that Home aggregate demand becomes more depressed in response to negative demand shocks than in a monetary union. When the Home ZLB binds, UIP dictates that the higher real rates in Home relative to Foreign real rates appreciate the Home currency and, hence, the terms of trade. This improvement in the terms of trade suppresses net exports in Home through expenditure switching away from Home goods. However, in a monetary union, the terms of trade barely affect net exports because they are a state variable and partly because prices are sticky, thus displaying sluggish dynamics. Thus, Home net exports are smaller under a flexible exchange rate regime than under a monetary union. Moreover, in a monetary union, a price-level targeting-like channel arises because the terms of trade are unaffected by nominal exchange rates and is purely determined by the ratio of the price of Foreign goods to the price of Home goods. In response to negative Home demand shocks, the price of Home goods initially falls, leading to depreciation of the terms of trade. To restore the terms of trade to its long-run level, the expected Home inflation must increase. This increase in expected future inflation alleviates the decline in consumption. The absence of the price-level targeting-like channel in a flexible exchange rate regime causes Home consumption to decrease more in a flexible exchange rate regime than in a monetary union. Therefore, the combination of smaller net exports and a larger decrease in consumption results in a larger aggregate demand contraction at the ZLB in a flexible exchange rate regime, motivating a higher optimal inflation target than in a monetary union. Due to the larger cost of price dispersion that stems from a higher optimal inflation target in a flexible exchange regime, a monetary union is better off than a flexible exchange regime in terms of welfare.

The second result concerns the optimal inflation target when UIP is violated. Departures from UIP are modeled by introducing an exogenous UIP shock in the UIP condition, as in [Itskhoki and Mukhin \(2021\)](#). The conclusion is that the optimal inflation target in a flexible exchange rate regime becomes higher than when UIP holds. If a violation of UIP is significant, the optimal inflation target in an open economy can be even larger than that in a closed economy. Moreover, the optimal inflation target in a flexible exchange rate regime relative to that in a monetary union increases as departures from UIP become more pronounced. Accordingly, in terms of welfare, a monetary union is more beneficial than a flexible exchange rate regime.

These results arise because of the two roles played by a UIP shock, which exerts further downward pressure on aggregate demand at the ZLB. First, it drives a wedge between terms of trade and relative real interest rates. If this wedge moves adversely, it exacerbates the perverse response of the terms of trade resulting from a rise in relative real rates. Consequently, more pronounced expenditure switching away from Home goods arises, depressing Home net exports. Second, the UIP shock acts as exogenous friction in international financial markets in the sense that it counteracts the savings flow from Home to Foreign, which is initiated by negative Home demand shocks. This friction makes it more difficult for savings to flow from Home to Foreign, generating more deflationary pressure than in the absence of a UIP shock. Thus, there is a greater increase in Home real rates and, consequently, a more significant contraction in Home consumption. The aggravated aggregate demand contraction at the ZLB warrants a higher optimal inflation rate in an open economy when UIP breaks than when it does not, enhancing the desirability of a monetary union.

Related literature. This paper is related to the literature on optimal trend inflation, which quantifies the trade-off between the benefits of avoiding the ZLB and the costs of positive steady-state inflation.² [Coibion et al. \(2012\)](#) and [Carreras et al. \(2016\)](#) derive the optimal inflation target, taking into account both the costs and benefits of positive trend inflation in a New Keynesian model with Calvo pricing. [Blanco \(2021\)](#) argues that the cost of positive steady-state inflation is much lower in the menu cost model, implying a higher optimal inflation target than in the model with Calvo pricing. Because these studies derive their results in a closed economy setting, they do not consider the role of trade openness. We demonstrate that this international factor eases aggregate demand contraction at the ZLB, thereby substantially lowering optimal trend inflation.³

² Some studies propose different mechanisms orthogonal to the ZLB concern to generate a positive optimal inflation target. [Kim and Ruge-Murcia \(2009\)](#) and [Mineyama \(2022\)](#) show that downwardly rigid nominal wages offer benefits for raising the optimal inflation rate. [Adam and Weber \(2019\)](#) demonstrate that, in a model incorporating heterogeneous firms and trends in firm-level productivity, optimal inflation might be positive even without any ZLB concern.

³ [Eggertsson and Krugman \(2012\)](#) and [Bhattarai et al. \(2018\)](#) discuss the paradox of flexibility in a closed economy, in which increased price flexibility causes more deflationary pressure, leading to greater aggregate demand contraction. Our finding that trade openness mitigates deflationary pressure for a given degree of price flexibility implies that trade openness can be viewed as one possible ingredient that alleviates the paradox of flexibility.

This paper also relates to the literature on optimal cooperative monetary policies in open economies when countries face sizable demand shocks that bring them to the ZLB. [Cook and Devereux \(2013\)](#) and [Fujiwara et al. \(2013\)](#) study optimal policy rates under a zero inflation target and show that a country's monetary policy rate should depend on another country's macroeconomic conditions when the goal is to maximize world welfare. By contrast, we allow policy rates to respond to domestic variables and find the welfare-maximizing inflation target, proposing an alternative policy tool by which central banks can coordinate to improve world welfare. While [Zhao \(2022\)](#) does incorporate positive trend inflation, she neither considers the ZLB nor investigates optimal trend inflation.

Finally, our paper contributes to the literature on the desirability of a monetary union. [Cook and Devereux \(2016\)](#) argue that, in response to negative demand shocks, a monetary union is beneficial over a flexible exchange rate regime only when the ZLB binds, as it can achieve reduced macroeconomic volatility due to the absence of a perverse response in the terms of trade. [Cho et al. \(2023\)](#) reach a similar conclusion in cases where the origin of asymmetric macroeconomic responses lies in varying degrees of price stickiness among different countries.⁴ They draw this conclusion assuming zero trend inflation, which implies no differences in the cost of steady-state inflation. By contrast, we compare the two exchange rate regimes under their respective optimal inflation targets and find that the cost of steady-state inflation plays an important role in determining welfare ranking. Despite the greater output volatility, a monetary union is beneficial due to its lower optimal inflation target. Thus, our contribution emphasizes a previously overlooked factor in determining welfare rank between exchange rate regimes: trend inflation.

The rest of this paper is structured as follows. Section 2 presents the basic model, derives the welfare loss function, and describes parameter values. Section 3 quantifies the optimal inflation targets and welfare for each exchange rate regime when UIP holds. Section 4 quantifies the optimal inflation targets and welfare when UIP breaks. Section 5 concludes.

2. A two-country model with trend inflation

We extend the two-country New Keynesian model described in [Corsetti et al. \(2011\)](#) and [Groll and Monacelli \(2020\)](#) by allowing for positive trend inflation. The world economy consists of two countries: Home and Foreign. The continuum of agents in Home and Foreign is populated at intervals $[0, n)$ and $[n, 1]$, respectively. The agents in each country are both consumers and monopolistic producers of differentiated goods. We assume that the producers set prices under producer currency pricing (PCP). The asset markets are complete, implying that UIP holds. For brevity, we describe only the decision problem for Home and mark the Foreign variables with asterisks. The problem for Foreign can be written symmetrically.

2.1. Households

A representative household in Home maximizes the present discounted value of lifetime utility:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left\{ \frac{Z_{H,t} C_t^{1-\sigma}}{1-\sigma} - \frac{1}{n} \frac{1}{1+\zeta} \int_0^n N_t(i)^{1+\zeta} di \right\} \quad (1)$$

subject to the budget constraint:

$$P_t C_t + \sum_{h^{t+1}} q_t(h^{t+1}) B_{t+1} + B_{H,t+1} \leq B_t + R_t B_{H,t} + \frac{1}{n} \int_0^n W_t(i) N_t(i) di + \tau_t, \quad (2)$$

where β is the discount factor, C_t denotes the consumption basket of Home priced at P_t , and $N_t(i)$ is Home's labor supply to produce good i . σ is the inverse of the intertemporal elasticity of substitution, and ζ is the inverse of Frisch elasticity of labor supply. B_{t+1} is the holdings of state-contingent securities, priced at $q_t(h^{t+1})$, paying off one unit of domestic currency when state h^{t+1} is realized at time $t+1$. $B_{H,t}$ is the domestic nominal bond, yielding a gross yield of R_t at the beginning of period t . τ_t is the lump-sum taxes. $Z_{H,t}$ represents the demand shock that follows the lognormal AR(1) process:

$$z_{j,t} = \rho_z z_{j,t-1} + \epsilon_{j,t}^z \quad \text{with} \quad \epsilon_{j,t}^z \stackrel{iid}{\sim} \mathcal{N}(0, \sigma_z^2) \quad (j = H, F), \quad (3)$$

where $z_{j,t} \equiv \log Z_{j,t}$.

The consumption basket of Home is a composite index of domestic and imported goods:

$$C_t = \left[(1-\gamma)^{\frac{1}{\eta}} C_{H,t}^{\frac{\eta-1}{\eta}} + \gamma^{\frac{1}{\eta}} C_{F,t}^{\frac{\eta-1}{\eta}} \right]^{\frac{\eta}{\eta-1}}, \quad (4)$$

where $\gamma = (1-n)\alpha$ is the weight of the imported goods in the consumption basket, with $\alpha \in [0, 1]$ denoting the degree of trade openness. $\eta > 0$ is the trade elasticity of substitution between domestic and foreign goods. $C_{H,t}$ and $C_{F,t}$ are consumption bundles of imperfectly substitutable varieties with the elasticity of substitution being ϵ :

$$C_{H,t} = \left[\left(\frac{1}{n} \right)^{\frac{1}{\epsilon}} \int_0^n C_{H,t}(i)^{\frac{\epsilon-1}{\epsilon}} di \right]^{\frac{\epsilon}{\epsilon-1}} \quad \text{and} \quad C_{F,t} = \left[\left(\frac{1}{1-n} \right)^{\frac{1}{\epsilon}} \int_n^1 C_{F,t}(i)^{\frac{\epsilon-1}{\epsilon}} di \right]^{\frac{\epsilon}{\epsilon-1}}, \quad (5)$$

⁴ [Groll and Monacelli \(2020\)](#) compare the impact of markup shocks in two exchange rate regimes under zero trend inflation and argue for the superiority of a monetary union due to the price-level targeting-like effect of the terms of trade.

where the demand for each variety is given by:

$$C_{H,t}(i) = \frac{1}{n} \left(\frac{P_{H,t}(i)}{P_{H,t}} \right)^{-\varepsilon} C_{H,t} \quad \text{and} \quad C_{F,t}(i) = \frac{1}{1-n} \left(\frac{P_{F,t}(i)}{P_{F,t}} \right)^{-\varepsilon} C_{F,t}. \quad (6)$$

$P_{j,t}(i)$ is the price of good i produced in country $j \in \{H, F\}$, quoted in the Home currency. The law of one price holds for each good, so $P_{j,t}(i) = S_t P_{j,t}^*(i)$, where S_t is the nominal exchange rate, which is the Home price of Foreign currency.

The aggregate price indices of $C_{H,t}$ and $C_{F,t}$ are:

$$P_{H,t} = \left[\left(\frac{1}{n} \right) \int_0^n P_{H,t}(i)^{1-\varepsilon} di \right]^{\frac{1}{1-\varepsilon}} \quad \text{and} \quad P_{F,t} = \left[\left(\frac{1}{1-n} \right) \int_n^1 P_{F,t}(i)^{1-\varepsilon} di \right]^{\frac{1}{1-\varepsilon}}, \quad (7)$$

while the price index of the consumption basket C_t is:

$$P_t = \left[(1-\gamma) P_{H,t}^{1-\eta} + \gamma P_{F,t}^{1-\eta} \right]^{\frac{1}{1-\eta}}. \quad (8)$$

The terms of trade are $T_t = \frac{P_{F,t}}{P_{H,t}} = \frac{S_t P_{F,t}^*}{P_{H,t}}$, and the real exchange rate is $Q_t = \frac{S_t P_t^*}{P_t}$.

2.2. Producers

Each firm i produces a differentiated good according to the production function:

$$Y_t(i) = A_{H,t} N_t(i). \quad (9)$$

Technology $A_{H,t}$ is common to all firms in each country, following the lognormal AR(1) process:

$$a_{j,t} = \rho_a a_{j,t-1} + \epsilon_{j,t}^a \quad \text{with} \quad \epsilon_{j,t}^a \stackrel{iid}{\sim} \mathcal{N}(0, \sigma_a^2) \quad (j = H, F), \quad (10)$$

where $a_{j,t} \equiv \log A_{j,t}$.

In a given period, a fraction $1 - \theta_H$ of producers choose the optimal reset price P_t^ρ to maximize the expected discounted sum of profits:

$$\max_{P_{H,t}^\rho(i)} \mathbb{E}_t \sum_{k=0}^{\infty} \theta_H^k v_{H,t,t+k} \left[(1 - \tau_{H,t+k}) P_{H,t}^\rho(i) Y_{t+k|t}(i) - W_{t+k}(i) N_{t+k|t}(i) \right] \quad (11)$$

subject to the isoelastic demand:

$$Y_{t+k|t}(i) = \left(\frac{P_t^\rho(i)}{P_{t+k}} \right)^{-\varepsilon} Y_{t+k}, \quad (12)$$

where $N_{t+k|t}(i)$ is the labor utilized at time $t+k$ conditional on the reset price at time t , and $v_{H,t,t+k} = \beta^k \frac{U_{c,t+k}}{U_c} \frac{P_t}{P_{t+k}}$ is the stochastic discount factor. The optimal reset price is:

$$P_{H,t}^\rho(i)^{1+\varepsilon\zeta} = \frac{\mathbb{E}_t \sum_{k=0}^{\infty} (\theta_H \beta)^k P_{H,t+k}^{\varepsilon(1+\zeta)} \left(\frac{Y_{t+k}}{A_{t+k}} \right)^{1+\zeta}}{\mathbb{E}_t \sum_{k=0}^{\infty} (\theta_H \beta)^k \frac{U_{c,t+k}}{P_{t+k}} \left(\frac{1}{\mathcal{M}_{H,t+k}} \right) P_{H,t+k}^\varepsilon Y_{t+k}}, \quad (13)$$

where $\mathcal{M}_{H,t+k} = \frac{\varepsilon}{(\varepsilon-1)(1-\tau_{H,t+k})}$ denotes the tax-adjusted markup as in [Groll and Monacelli \(2020\)](#). We assume that there are no distortions due to market power in the steady state by letting $\tau_H = -\frac{1}{\varepsilon-1}$. The markup evolves exogenously due to the time-varying tax on producers, following the lognormal AR(1) process:

$$\mu_{j,t} = \rho_\mu \mu_{j,t-1} + \epsilon_{j,t}^\mu \quad \text{with} \quad \epsilon_{j,t}^\mu \stackrel{iid}{\sim} \mathcal{N}(0, \sigma_\mu^2) \quad (j = H, F), \quad (14)$$

where $\mu_{j,t} \equiv \log \mathcal{M}_{H,t+k}$.

2.3. Monetary policy

We assume that the steady-state gross inflation is the same in both countries (i.e., $\Pi = \Pi_H = \Pi_F$). In a flexible exchange rate regime, the central bank in each country sets the policy rate according to the Taylor rule, which responds to deviations from country-specific inflation:

$$R_t = \max \left[\frac{\Pi}{\beta} \left(\frac{\Pi_{H,t}}{\Pi} \right)^{\phi_\pi}, 1 \right]; \quad R_t^* = \max \left[\frac{\Pi}{\beta} \left(\frac{\Pi_{F,t}^*}{\Pi} \right)^{\phi_\pi}, 1 \right], \quad (15)$$

where $\Pi_{H,t} = \frac{P_{H,t}}{P_{H,t-1}}$ and $\Pi_{F,t}^* = \frac{P_{F,t}^*}{P_{F,t-1}^*}$. ϕ_π captures the strength of the policy response. In a monetary union, the union-wide central bank responds to deviations in the average of the two countries' inflation:

$$R_t = \max \left[\frac{\Pi}{\beta} \left(\frac{\Pi_{H,t}}{\Pi} \right)^{n\phi_\pi} \left(\frac{\Pi_{F,t}^*}{\Pi} \right)^{(1-n)\phi_\pi}, 1 \right]. \quad (16)$$

2.4. Solution method

As in Coibion et al. (2012), we log-linearize the model around the steady state in which inflation need not be zero. Define $\tilde{\omega}_t \equiv \log\left(\frac{\omega_t}{\omega_t^f}\right)$ and $\hat{\omega}_t \equiv \log\left(\frac{\omega_t}{\omega}\right)$, where ω_t^f is the efficient level of ω_t and ω is the nonstochastic steady-state level of ω_t .⁵ Using these notations, we next describe the welfare metric of the world economy that is used to compute the optimal steady-state inflation.

2.5. Welfare function

We assume that the expected per-period world welfare is given by the weighted average of the expected per-period utilities of the two countries and find a common inflation target that maximizes expected per-period world welfare. Specifically, following the approach of Coibion et al. (2012), we derive a second-order approximation of expected per-period world welfare, which is:

$$\Theta_0 - \frac{\sigma + \zeta}{2} \left[\text{var}(\tilde{y}_t^W) + n(1-n)\text{var}(\tilde{y}_t^R) \right] + \Theta_1 \text{var}(\tilde{T}_t) + \Theta_2 \left[\text{var}(\hat{\pi}_t^W) + n(1-n)\text{var}(\hat{\pi}_t^R) \right], \quad (17)$$

where $\tilde{y}_t^W \equiv n\tilde{Y}_t + (1-n)\tilde{Y}_t^*$, $\tilde{y}_t^R \equiv \tilde{Y}_t - \tilde{Y}_t^*$, $\hat{\pi}_t^W \equiv n\hat{\pi}_{H,t} + (1-n)\hat{\pi}_{F,t}^*$, $\hat{\pi}_t^R \equiv \hat{\pi}_{H,t} - \hat{\pi}_{F,t}^*$, and $\tilde{T}_t$ denote the world average output gap, the relative output gap, world average inflation, relative inflation, and the terms of trade gap, respectively. The Online Appendix provides the derivation of Eq. (17) and coefficients Θ_0 , Θ_1 , and Θ_2 .

As identified by Coibion et al. (2012), the constant term $\Theta_0 \leq 0$ captures the welfare loss that arises from positive steady-state inflation. The higher the steady-state inflation, the more dispersed the steady-state prices across firms, leading to a more inefficient allocation of resources. This welfare cost is present in both countries, and the size of the cost is identical because of symmetric steady-state inflation and price stickiness. $\Theta_2 < 0$ represents the sensitivity of the welfare loss to the variation of world and relative inflation. The higher the level of trend inflation, the larger the welfare cost of a given amount of inflation volatility. Because positive steady-state inflation already generates an inefficient allocation of workers across firms due to price dispersion, a country-specific inflationary shock becomes more costly as additional price dispersion leads to a more dispersed marginal disutility of labor. Given that marginal disutility is convex, a more dispersed marginal disutility leads to greater welfare loss. Lastly, although not explicitly captured in the welfare loss function, trend inflation increases the volatility of inflation. In particular, greater steady-state inflation induces more forward-looking behavior for firms that are able to reset their prices. Thus, a country-specific inflationary shock makes these firms set higher prices when steady-state inflation is high, leading to more volatile relative and world inflation.

In an open economy, the volatility of the terms of trade gap enters the welfare criterion, as documented by Pappa (2004), Corsetti et al. (2011), and Groll and Monacelli (2020), among many others. The terms of trade gap measures the extent to which resources are misallocated between countries due to the difference in the prices of final goods produced in the two countries. The coefficient on the variance of the terms of trade gap Θ_1 does not depend on trend inflation. However, this does not imply that positive steady-state inflation does not impose any welfare cost through international relative prices. Rather, as we will see, the dynamics of the terms of trade are significantly affected by the inclusion of the ZLB, which depends crucially on the level of trend inflation. The terms of trade dynamics are crucial for understanding relative aggregate demand, which drives the volatility of the relative and world variables. In this sense, trend inflation does affect welfare losses through the terms of trade gap.

2.6. Calibration

We treat Home as the group of core countries in the E.U. and Foreign as the group of periphery countries. Specifically, we define the Euro area core countries as Austria, Belgium, Finland, France, Germany, and the Netherlands, following the classification by Gilchrist et al. (2023). The remaining 21 countries are considered as the periphery.⁶

We target the relative size of Home n to match the share of the core GDP in the Euro area's GDP, which is 0.55 as of 2022. By rounding up this value, we obtain $n = 0.6$. The weight of the imported goods in the consumption basket in Home $\gamma = (1-n)\alpha$ is set to 0.2, consistent with the average import dependence of final consumption expenditure in Germany between 2010–2020.⁷ The values of n and γ imply that the weight of imported goods in the consumption basket in Foreign is $\gamma^* = n\alpha = 0.3$. The discount factor β is 0.9961, which implies an annual real interest rate of 1.57%, the average real rate in Germany between 1970 and 2023. The inverse elasticity of intertemporal substitution σ is 1. The inverse of Frisch elasticity of labor supply ζ is 1. The intratemporal trade elasticity η , or the elasticity of substitution between goods across countries, is set to 2, which is in an empirically plausible range estimated by Lubik and Schorfheide (2005), Broda and Weinstein (2006), and Simonovska and Waugh (2014).⁸

Since the optimal inflation rate is the point at which the marginal cost and benefit of increasing trend inflation are equal, it is necessary to consider the parameters that govern these costs and benefits. Parameters that are important for the steady-state

⁵ The efficient level of ω_t is the value of ω_t that prevails under flexible prices and without distortions from monopolistic competition.

⁶ These countries are Bulgaria, Croatia, Republic of Cyprus, Czech Republic, Denmark, Estonia, Greece, Hungary, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Poland, Portugal, Romania, Slovakia, Slovenia, Spain and Sweden.

⁷ Since Germany is the largest country in terms of GDP in the core group, we assume that Germany is a representative country of the core group.

⁸ Numerous contributions that study optimal monetary policy in an open-economy New Keynesian model often assume $\eta > 1$, and thus we choose $\eta = 2$ to be a baseline value. Since the empirical literature provides a broad range of estimates, from 0.43 to 6, we provide the sensitivity of our results by considering cases $\sigma\eta < 1$, $\sigma\eta = 1$, and $\sigma\eta > 1$ in Appendix A.

Table 1
Baseline calibration.

Parameter	Value	Description
A. Structural parameters		
β	0.9961	Discount factor
σ	1	Inverse of the elasticity of intertemporal substitution
ζ	1	Inverse of Frisch elasticity of labor supply
ε	7	Elasticity of substitution between differentiated goods
θ_j	0.55	Calvo parameter
n	0.6	Relative size of Home
η	2	Elasticity of substitution between goods across countries
ϕ_π	1.96	Long-run response to inflation
B. Shock parameters		
ρ_z	0.84	Persistence of demand shocks
ρ_a	0.82	Persistence of technology shocks
ρ_μ	0	Persistence of cost-push shocks
$\sigma_{z,j}$	0.066	Standard deviation of demand shocks
$\sigma_{a,j}$	0.0062	Standard deviation of technology shocks
$\sigma_{\mu,j}$	0.0016	Standard deviation of cost-push shocks

Note: j is the subscript for a country, where $j \in \{H, F\}$.

cost of price dispersion include the elasticity of substitution across intermediate goods ε and the Calvo parameter θ_j .⁹ We set the elasticity of substitution across intermediate goods to 7, at which the steady-state markup is 17%, which is in line with the estimates provided by Basu and Fernald (1997). The Calvo parameter is set to 0.55 so that the average price duration is 7 months. This value is a middle value between the micro estimates of Bils and Klenow (2004), who report an average price duration of 4–5 months, and those of Nakamura and Steinsson (2008), who report an average price duration of 9–11 months.

The benefits of increasing inflation depend substantially on the severity of the recession at the ZLB and the frequency of hitting the ZLB. The response of monetary policy to inflation and output gap variability are instrumental parameters for these benefits. The stronger the systematic response of monetary policy to inflation and the output gap, the more it mitigates the deflationary spiral for a given duration of zero bound episodes, thus creating a smaller incentive to raise the inflation target. To be consistent with the European Central Bank (ECB)'s single mandate of price stability, we let the baseline monetary policy rule respond only to inflation and set the inflation reaction coefficient ϕ_π equal to 1.96. This value is from Hohberger et al. (2019), who estimate a DSGE model for the Euro area using data between 1991q1 and 2018q4.

The calibration of shock processes is crucial in determining the model's frequency of encountering the ZLB. The persistence and standard deviation of shocks are from Smets and Wouters (2003), who estimated shock processes for the Euro area, with the exception of the standard deviation of demand shocks. Regarding the size of the demand shocks $\sigma_{z,H}$ and $\sigma_{z,F}$, we choose their values so that our monetary union model predicts the historical frequency of hitting the ZLB in the Euro area under a steady-state inflation rate of 3.5%. However, due to the implementation of a negative deposit facility rate by the ECB in June 2014, there is a debate over whether a zero interest rate is the effective lower bound. To provide a conservative quantitative assessment of the optimal inflation rates and welfare in a flexible exchange rate regime and monetary union, we assume that the effective lower bound in the Euro area is -0.5% – -0.4% , the lowest values in the history of the E.U., prevailing during 2016Q2–2022Q2. We consider the sample period for short-term interest rates from 1970Q1, during which data for both Germany and France are available, until 2023Q4. Given this sample period, the frequency of hitting the effective lower bound is 12%, which we treat as our baseline ZLB frequency.¹⁰ It turns out that the persistence of the demand shocks ρ_z , technology shocks ρ_a , and cost-push shocks ρ_μ are 0.84, 0.82, and 0, respectively. The standard deviations of the demand shocks $\sigma_{z,j}$, technology shocks $\sigma_{a,j}$, and cost-push shocks $\sigma_{\mu,j}$ are 0.066, 0.0062, and 0.0016, respectively, for both countries.

Considering that the benefits of trade openness decline with higher correlations of demand shocks because the partner country provides less insurance, it is important to allow for a realistic cross-country correlation structure. We set the demand shock correlation as $\text{corr}(e_{H,t}^z, e_{F,t}^z) = 0.2$. This is a mid-value of the estimates provided by Bayoumi and Eichengreen (1993) and Campos and Macchiarelli (2016), who find that the demand shock correlation between Germany and countries in the periphery lies between -0.08 and 0.34 . The baseline parameter values used for the numerical simulation are summarized in Table 1.

⁹ We assume fixed price stickiness, which may be a strong assumption, given the recent evidence provided by L'Huillier and Schoenle (2023). They document a positive relationship between price flexibility and inflation targets. Even if we incorporate this relationship into our model, we still expect an open economy to deliver less contracted aggregate demand and, thus, a lower optimal inflation target than a closed economy. See Appendix C for the mechanism when price flexibility correlates with inflation targets.

¹⁰ The ZLB frequency chosen by Coibion et al. (2012) is 5% because they end their sample period at the end of 2011. However, U.S. interest rates remained at the ZLB until the end of 2015 and reached again during the COVID-19 recession. If the sample were extended to include data until 2023, the ZLB frequency would be much higher and closer to ours.

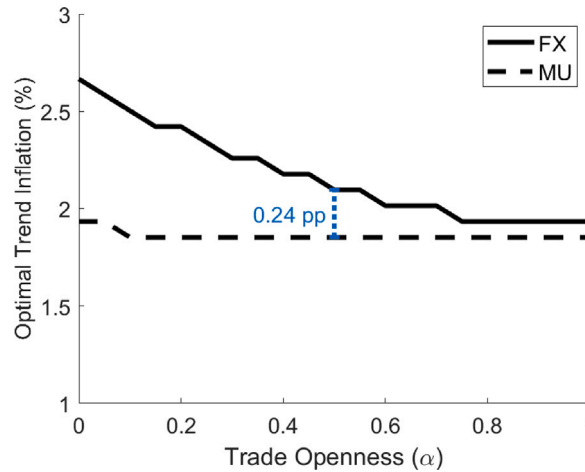


Fig. 1. Optimal trend inflation by trade openness.

Notes. The figure displays the optimal annual inflation rate under a flexible exchange rate regime and a monetary union for different degrees of trade openness α .

3. Optimal trend inflation when UIP holds

The goals of this section are twofold. First, we investigate whether the optimal inflation rate in a closed economy differs from that in an open economy in which UIP holds. To do so, we compute optimal trend inflation in a flexible exchange rate regime for different levels of trade openness. Second, we compare the optimal inflation rate in a flexible exchange rate regime with that in a monetary union at each level of trade openness. This comparison is important because the difference in optimal inflation rates affects the welfare rank between a flexible exchange rate regime and a monetary union and, thus, the desirability of forming a monetary union. To find the optimal trend inflation, we simulate the model for 100,000 periods for each steady-state inflation rate and search for the steady-state inflation rate that yields the smallest welfare loss. We utilize an algorithm developed by Holden (2016, 2023) that solves for the perfect foresight solution of linear models with occasionally binding constraints.

3.1. Flexible exchange rate regime

The solid line in Fig. 1 shows the optimal inflation target for different levels of trade openness under a flexible exchange rate regime. In the no-trade case ($\alpha = 0$), the closed economy studied by Coibion et al. (2012), the optimal inflation target is 2.67%. As trade openness increases, the optimal inflation rate decreases, indicating that an open economy delivers a lower optimal inflation rate than a closed economy.

The decreasing optimal inflation rate with trade openness arises because the costs of increasing trend inflation remain constant regardless of trade openness, whereas its benefits decrease with trade openness. As documented by Coibion et al. (2012), the costs of increasing trend inflation emerge mostly from the steady-state costs of positive inflation. Because steady-state world average inflation is the same regardless of trade openness, the steady-state cost of positive world inflation is identical across various levels of trade openness. The invariance of the steady-state cost is captured by Θ_0 in the welfare loss function (17), which is independent of trade openness α . The benefits of increasing trend inflation are captured by the extent to which the aggregate demand stabilizes when the inflation target is increased. Consider the aggregate demand contraction induced by negative Home demand shocks that cause the Home ZLB to bind. If trend inflation increases, the Home ZLB becomes slack, and aggregate demand contraction becomes less severe. As deflationary pressure in Home is alleviated, the volatility of the relative and world variables in the loss function (17) decreases, thereby improving welfare.

To understand the decreasing benefits of increasing trend inflation with trade openness, we consider a sequence of negative Home demand shocks that cause the Home ZLB to bind for 10 periods when trade openness α is zero, and trend inflation is zero. The first row of Fig. 2 shows the real interest rates, aggregate demand, and its components on impact in response to these shocks for different levels of trade openness, assuming zero trend inflation. The second row plots the impact response of the same variables to the same shocks when trend inflation is increased to 4%, under which the ZLB does not bind. Once we compare the right panels of rows 1 and 2, it is evident that the stabilization of Home aggregate demand arises most strikingly when trade openness α is zero and decreases with trade openness. The different degrees of aggregate demand stabilization across trade openness arise mostly from consumption and partly from net exports.

Why does consumption stabilization increase most significantly when trade is closed and gradually decrease as trade openness increases? In a closed economy setting, excess savings triggered by negative Home demand shocks generate deflationary pressure, which increases domestic real interest rates when the ZLB on nominal interest rates binds. The increased real rates put downward pressure on consumption, thereby contracting the aggregate demand in Home. When trade is open, part of Home's excess savings

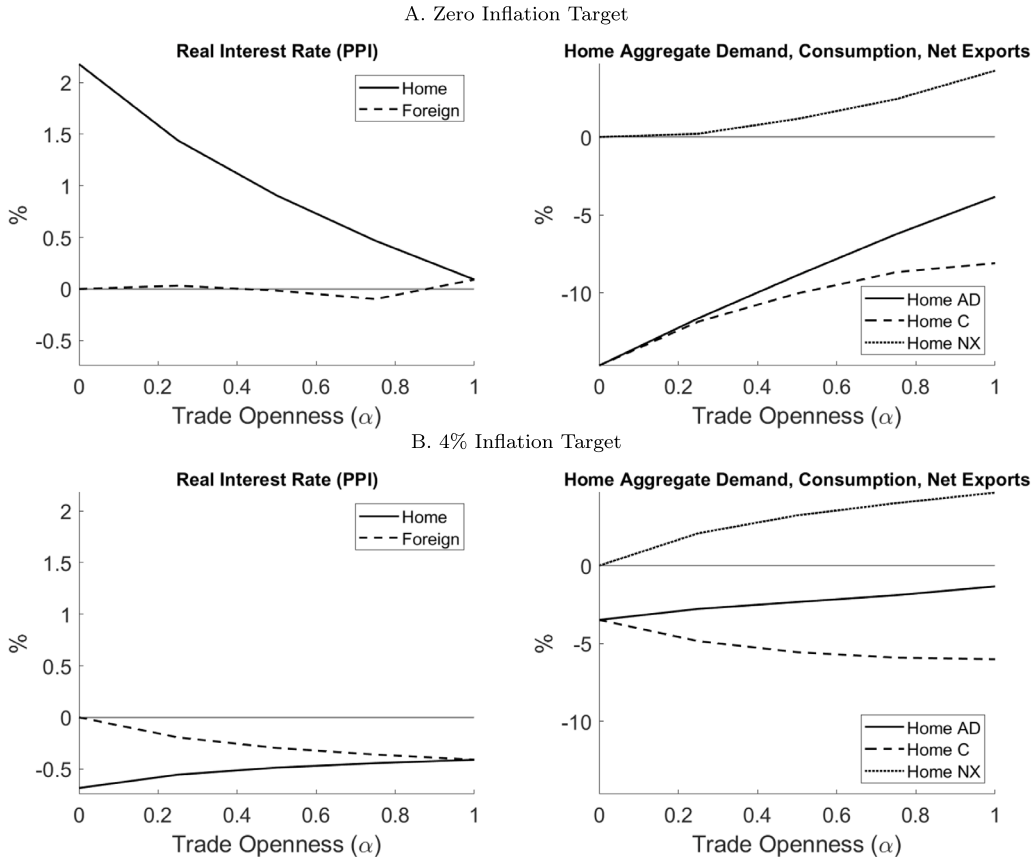


Fig. 2. Benefits of trend inflation in FX.

Notes. The first row shows the impact response of real interest rates in Home and Foreign, aggregate demand, and its components in Home to negative demand shocks that cause the Home ZLB to bind for 10 periods when the trend inflation is zero. The second row plots the impact response of the same variables to the same shocks when trend inflation is increased to 4%, under which the ZLB does not bind.

flow to Foreign through net exports, resulting in a smaller decline in output than in a closed economy. As deflationary pressure is less severe due to increased trade openness, real interest rates increase less at the ZLB, alleviating the decline in domestic consumption. In contrast, when the ZLB does not bind, the effect of the mitigated deflationary pressure on real interest rates works in the opposite direction due to an unconstrained Taylor rule. That is, the less severe deflationary pressure arising from increased trade openness causes real interest rates to fall less, thereby leading to more depressed consumption, as observed in row 2. The contrasting relationship between consumption and trade openness at and off the ZLB explains why aggregate demand stabilization decreases with trade openness upon increasing the inflation target.

A substantial reduction in Home real rates upon increasing trend inflation boosts not only consumption but also Home net exports. To understand the connection between real interest rates and net exports, we solve the log-linearized UIP condition forward, which yields:

$$\hat{T}_t = -\mathbb{E}_t \sum_{j=0}^{\infty} \left[\left(\hat{R}_{t+j} - \hat{\pi}_{H,t+j+1} \right) - \left(\hat{R}_{t+j}^* - \hat{\pi}_{F,t+j+1}^* \right) \right]. \quad (18)$$

According to this expression, an increase in relative real interest rates – the difference between the Home and Foreign real rates – causes a perverse movement of the terms of trade (i.e., a fall in $\hat{T}_t$), making Home goods relatively more expensive than Foreign goods. Consequently, expenditure switching away from Home goods arises, suppressing Home net exports. When the relative real rates fall upon raising the inflation target, as noted in the left panel of row 2 of Fig. 2, the terms of trade depreciate, generating expenditure switching toward Home goods. This increases Home net exports and boosts Home aggregate demand. As trade openness increases, the increase in relative real rates at the ZLB and the fall in relative real rates off the ZLB become less pronounced, so the extent to which Home net exports increase upon the rise in trend inflation declines. Thus, the less expansionary net exports associated with increased trade openness partly contribute to the decreasing benefits of increasing inflation targets.

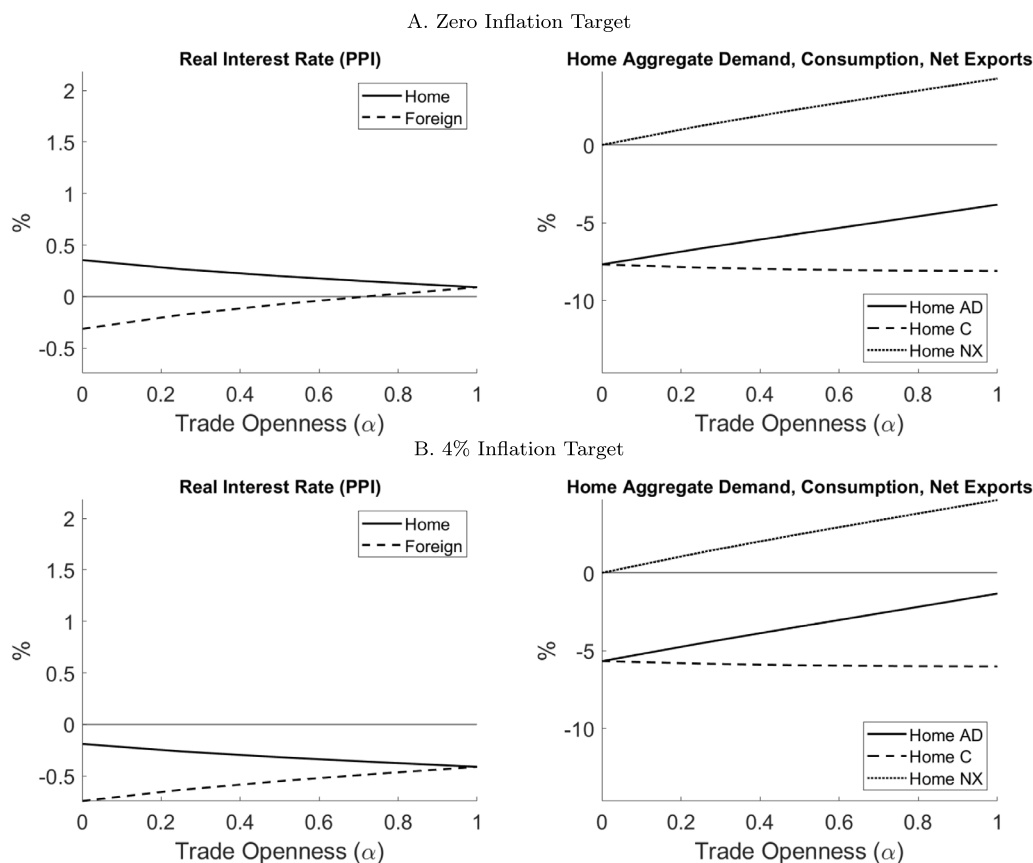


Fig. 3. Benefits of trend inflation in MU.

Notes. The first row shows the impact response of real interest rates in Home and Foreign, aggregate demand, and its components in Home to negative Home demand shocks, which we use to plot Fig. 2, when trend inflation is zero in a monetary union. The second row plots the same variables in response to these shocks when trend inflation increases to 4%, under which the ZLB does not bind.

In sum, while the costs of increasing the inflation target are insensitive to trade openness because its benefits decrease with trade openness, the optimal inflation target in a flexible exchange rate regime decreases with trade openness.¹¹

3.2. Monetary union

The dashed lines in Fig. 1 correspond to optimal inflation targets under a monetary union. Unlike a flexible exchange rate regime, the optimal inflation rate in a monetary union is nearly identical for all degrees of trade openness. The reasons for this are as follows. First, the constant costs of increasing trend inflation over trade openness that we discussed under a flexible exchange rate regime also apply to a monetary union. Second, the benefits of increasing trend inflation are nearly invariant in a monetary union.

To demonstrate the invariance of the benefits, we compare the aggregate demand under zero trend inflation and that under 4% trend inflation in a monetary union. The first row of Fig. 3 shows the real interest rates, aggregate demand, and its components in a monetary union on impact in response to negative Home demand shocks, which were used to plot Fig. 2, when the trend inflation is zero. The second row plots the same variables in response to these shocks when trend inflation increases to 4%, under which the ZLB does not bind. The difference between aggregate demand under zero and a 4% inflation target captures the benefits of increasing the inflation target by 4% in a monetary union. Comparing the right panels of rows 1 and 2, we find that the stabilization of Home aggregate demand is nearly identical across trade openness α . The near-identical stabilization is due to unchanged net exports and a parallel shift in the flat consumption response across degrees of trade openness.

¹¹ Because our model does not incorporate a nontradable sector, our results differ from what Krugman (1998) predicts. Krugman (1998) argues that exporting savings in the form of net exports may not be a realistic mechanism to alleviate the repercussions of the ZLB. This is because, in large economies like Japan or the United States, the bulk of goods and services are nontradable, limiting the flow of capital from the more affected country to the less affected one. If we incorporate an empirically relevant nontradable sector, savings through exports might be much smaller than what our current model predicts. In this scenario, we would still expect the model with a large nontradable share to result in a lower optimal inflation target in an open economy than in a closed economy, but to a lesser extent.

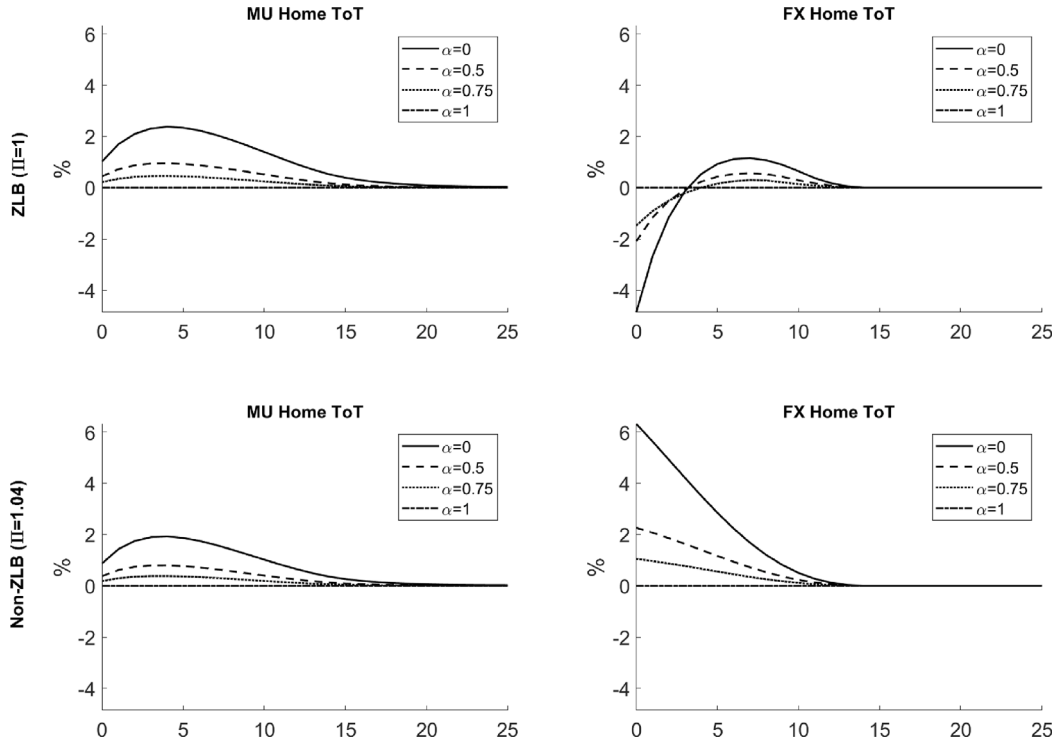


Fig. 4. Terms of trade: MU vs FX.

Notes. The figure shows the impulse responses of the terms of trade in a monetary union (MU) and a flexible exchange regime (FX) to negative Home demand shocks, which were used to plot Fig. 2, for different levels of trade openness α . The first row corresponds to a scenario of zero trend inflation, under which the ZLB binds, and the second row corresponds to a scenario of 4% trend inflation, under which the ZLB does not bind.

3.2.1. Unchanged net exports

Home net exports remain unchanged when the inflation target is increased to 4% because of the sluggish dynamics of terms of trade in a monetary union. The terms of trade dynamics are sluggish because the terms of trade are a state variable in a monetary union and partly because prices are sticky. Because the terms of trade are insulated from nominal exchange rate fluctuations, the log-linearized terms of trade dynamics in a monetary union become:

$$\hat{T}_t = \hat{\pi}_{F,t}^* - \hat{\pi}_{H,t} + \hat{T}_{t-1}. \quad (19)$$

This specification holds both at and off the ZLB, indicating that the terms of trade are a state variable. As a result, unlike under flexible exchange rates, the terms of trade at the ZLB shows nearly identical movements as those away from the ZLB, exhibiting sluggish dynamics. Fig. 4 depicts the impulse responses of the terms of trade to negative demand shocks in a monetary union compared to those in a flexible exchange rate regime. Again, these shocks are used to plot Fig. 2. The left panels of the figure highlight the near equivalence of the terms of trade at the ZLB and that away from the ZLB in a monetary union for a given trade openness α . However, under flexible exchange rates, except when real rates are the same across countries ($\alpha = 1$), the sign of the terms of trade dynamics at the ZLB is completely different from that away from the ZLB. At the ZLB, the terms of trade appreciate because Home real rates are larger than Foreign real rates, while it depreciates away from the ZLB because Home real rates are lower than Foreign real rates.

The near equivalence of the terms of trade at and away from the ZLB in a monetary union implies that Home net exports are nearly identical at and away from the ZLB. This point can be seen in log-linearized Home net exports¹²:

$$n\hat{x}_t = \frac{\alpha}{2\sigma} [(2 - \alpha)(\sigma\eta - 1) + (1 - \sigma)]\hat{T}_t + \frac{\alpha}{2\sigma} (z_{F,t} - z_{H,t}). \quad (20)$$

Home net exports are determined by the terms of trade response and the direct effect of relative demand shocks, where the latter effect arises independently of monetary policy. As the terms of trade are nearly identical at and away from the ZLB in a monetary union, so are the Home net exports. Therefore, even when the ZLB no longer binds due to an increase in the inflation target, the response of net exports to the given Home demand shocks remains essentially the same in a monetary union.

¹² See the Online Appendix for the derivation.

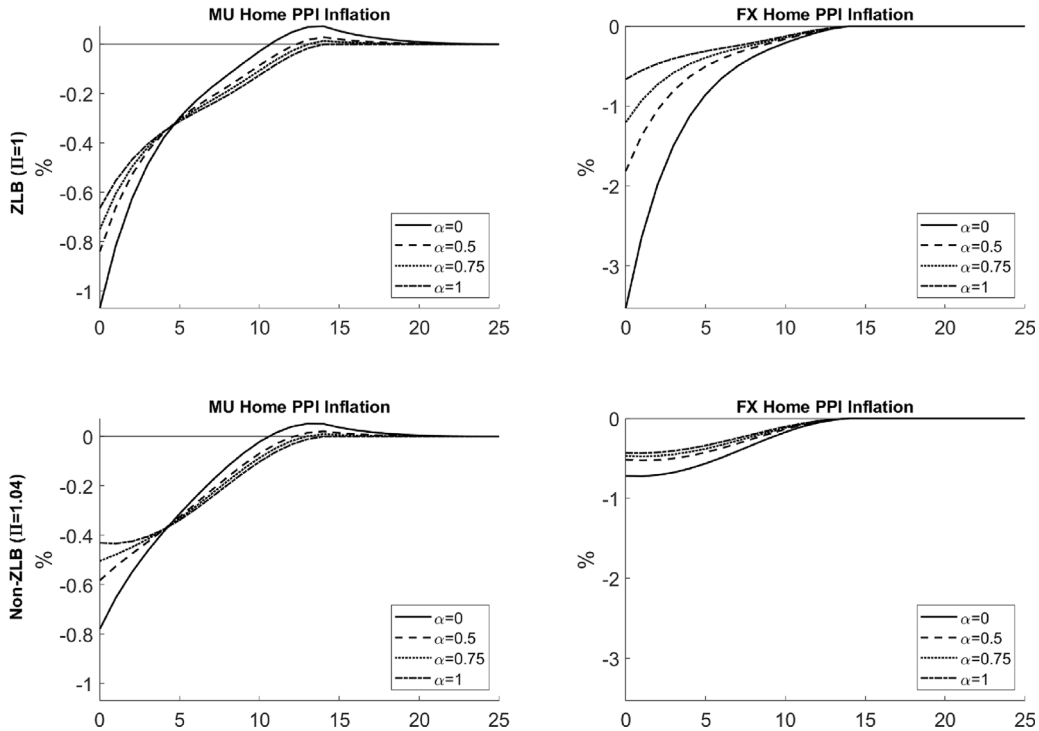


Fig. 5. Home inflation: MU vs FX.

Notes. The figure shows the impulse responses of Home inflation in a monetary union (MU) and a flexible exchange regime (FX) to negative Home demand shocks, which were used to plot Fig. 2, for different levels of trade openness α . The first row corresponds to a scenario of zero trend inflation, under which the ZLB binds, and the second row corresponds to a scenario of 4% trend inflation, under which the ZLB does not bind.

3.2.2. Parallel shift in the flat consumption response

Notice from Fig. 3 that when trend inflation is zero, and the ZLB binds, the consumption responses are nearly flat with respect to trade openness. When trend inflation is increased to 4%, and the ZLB does not bind, the consumption response shifts up parallelly, maintaining a flat shape over trade openness. The reason consumption responses are near-flat across different levels of trade openness arises from two offsetting forces. On the one hand, as trade openness increases, reduced deflationary pressure causes Home real rates to increase less, causing consumption to show an increasing pattern over trade openness. On the other hand, as trade openness increases, the weakening price-level targeting-like channel limits consumption stabilization, resulting in a decreasing pattern of consumption over trade openness. The two offsetting forces are present both at and away from the ZLB. Below, we elucidate these two forces, starting with the case at the ZLB.

At the ZLB. Recall, under flexible exchange rates, trade openness subdues the rise in real interest rates at the ZLB, leading to a less severe consumption fall. This channel is also operative under a monetary union. If this were the only channel, the consumption response would exhibit an increasing pattern over trade openness. However, in a monetary union, there is a price-level targeting-like channel that mitigates the aggregate demand contraction.

As discussed in Groll and Monacelli (2020), the price-level targeting-like channel arises because, in a monetary union, the terms of trade are unaffected by the nominal exchange rates and is purely determined by the ratio of the price of Foreign goods to the price of Home goods, $T_t = \frac{P_{F,t}^e}{P_{H,t}^e}$. In response to negative Home demand shocks, the price of Home goods falls on impact, resulting in the depreciation of terms of trade. To ensure that the terms of trade converge to its long-run level, the expected Home inflation must rise. The effect of an increase in expected future inflation works similarly to the price-level targeting policy discussed in the closed economy New Keynesian literature (Eggertsson and Woodford, 2003). That is, it alleviates the severity of economic contraction in the current period by causing agents to expect a boom in the future. The price-level targeting-like channel arises most powerfully when trade openness α is zero and gradually weakens as trade openness increases. This is because the lower the trade openness, the stronger the deflationary pressure in Home, leading to a substantial fall in Home goods prices. Recovery to the long-run level of terms of trade requires higher expected future Home inflation, counteracting the aggregate demand contraction today to a larger extent. If the price-level targeting-like channel were the only one, the consumption response would exhibit a decreasing pattern over trade openness.

The upper-left panel of Fig. 5 provides a visual inspection of the price-level targeting-like effect of the terms of trade in a monetary union at the ZLB. It illustrates the impulse responses of Home inflation to negative Home demand shocks for different levels of trade openness. Note that the lower the trade openness, the greater the decrease in Home inflation on impact. The greater fall of Home

inflation on impact entails more positive future inflation to ensure that the terms of trade return to its steady-state level. As trade openness increases, the impact fall of Home inflation becomes smaller, implying a smaller deviation in the terms of trade from the steady state. In this scenario, the required future inflation to achieve steady-state terms of trade is lower, indicating a weakened price-level targeting-like effect. However, as shown in the upper-right panel, positive future inflation does not arise under flexible exchange rates. Because the terms of trade deviation that occurred from the fall in domestic prices is made up by the nominal exchange rate appreciation, there is no need to generate positive expected future inflation to attain steady-state terms of trade in the long run.

Away from the ZLB. Notice from Fig. 3, Home real interest rates shift down for all levels of trade openness upon a rise in the inflation target, as the ZLB no longer binds. This reduction in Home real rates boosts consumption at all levels of trade openness. However, as in the case of the ZLB, the consumption responses are nearly flat across different levels of trade openness, indicating a parallel upward shift in these responses upon raising the inflation target. Flat consumption responses result from two offsetting channels, which are described below.

In a monetary union, even if monetary policy is unconstrained by the ZLB, nominal rates cannot fall sufficiently in response to contractionary Home demand shocks because the policy rates react to world average inflation, which does not decline as much as domestic inflation. Consequently, the larger the decline in Home inflation, the greater the Home real rates. Since the decline in Home inflation is the largest when trade openness α is zero, Home real rates are the highest as well, as observed in the bottom-left panel of Fig. 3. As trade openness increases, Home real rates decrease, owing to reduced deflationary pressure. If this were the only channel, the domestic consumption response would exhibit an increasing pattern over trade openness.

The price-level targeting-like channel in a monetary union is still operative even if the ZLB does not bind. Again, this channel arises from the definition of terms of trade unaffected by nominal exchange rates, which applies both at and away from the ZLB. Hence, the presence of this channel is independent of whether the economy is at the ZLB. The bottom-left panel of Fig. 5 visualizes the existence of a price-level targeting-like channel under a 4% inflation target, where the ZLB does not bind, by showing that current deflation is associated with future inflation. This figure also illustrates that the channel weakens as trade openness increases. Again, if the price-level targeting-like channel was the only channel, the domestic consumption response would exhibit a declining pattern over trade openness.

In sum, in a monetary union, aggregate demand stabilization upon raising the inflation target is near-invariant with respect to trade openness because net exports are barely affected, and consumption rises equally across all levels of trade openness. Since both costs and benefits are insensitive to trade openness, the optimal inflation target in a monetary union does not change across different levels of trade openness.

3.3. Quantitative assessment: FX vs. MU

Having discussed why the patterns of optimal inflation rates over trade openness are different between a flexible exchange rate regime and a monetary union, we now provide a direct comparison of the optimal inflation rates between the two exchange rate regimes – a flexible exchange rate regime and a monetary union – under our E.U. calibration. Recall that the weight of imported goods in the consumption basket in Home γ is 0.2, and the relative size of Home n is 0.6. These values imply trade openness α of 0.5, under which the optimal inflation rate is 2.09% in a flexible exchange rate regime and 1.85% in a monetary union.

The higher optimal inflation rate in a flexible exchange rate regime stems from Home aggregate demand being more depressed than in a monetary union at the ZLB. Note that Eq. (18), which emerges from the UIP condition, applies only under flexible exchange rates and links the terms of trade response to the relative real interest rates. When the Home ZLB binds, the higher real rates in Home relative to Foreign real rates appreciate the Home currency and, hence, the terms of trade. This improvement in the terms of trade suppresses net exports in Home through expenditure switching away from Home goods. However, in a monetary union, the terms of trade barely affect net exports because it is a state variable that displays a sluggish movement. Thus, Home net exports are lower under a flexible exchange rate than under a monetary union. Moreover, the absence of a price-level targeting-like channel causes Home consumption to decrease more in a flexible exchange rate regime than in a monetary union. Therefore, the combination of smaller net exports and a larger decrease in consumption results in a larger aggregate demand contraction at the ZLB in a flexible exchange rate regime, explaining the higher optimal inflation target than in a monetary union.^{13,14}

¹³ In Appendix A, we discuss the sensitivity of optimal inflation rates with respect to alternative trade elasticities. Trade elasticity does not matter for the optimal inflation rates in either exchange rate regime when UIP holds in the sense that they are virtually the same across different values of trade elasticity. Appendix B provides a sensitivity analysis with respect to different monetary policy rules. It shows that the optimal inflation rate results in this section remain robust to the output gap targeting rule in that optimal inflation rates decrease with trade openness and are higher in a flexible exchange rate regime than in a monetary union. However, when the monetary policy rule targets the price level, the optimal inflation rates in the two exchange rate regimes are less than 0.3% regardless of trade openness.

¹⁴ Although the producer currency pricing (PCP) assumption is commonly made, there are works that adopt the local currency pricing (LCP) assumption, under which export prices are set in the destination's currency (Engel, 2011; Corsetti et al., 2023). Under LCP, we expect a flexible exchange rate regime to deliver a lower optimal trend inflation rate than under PCP. Thus, the welfare gain of forming a monetary union may become smaller. The reason is that, under LCP, the exchange rate pass-through is less pronounced than under PCP, suggesting that the terms of trade are insulated from perverse exchange rate fluctuations. Consequently, the cost of the ZLB stemming from perverse terms of trade movements under a flexible exchange rate regime decreases. We leave the quantitative analysis under LCP for future work.

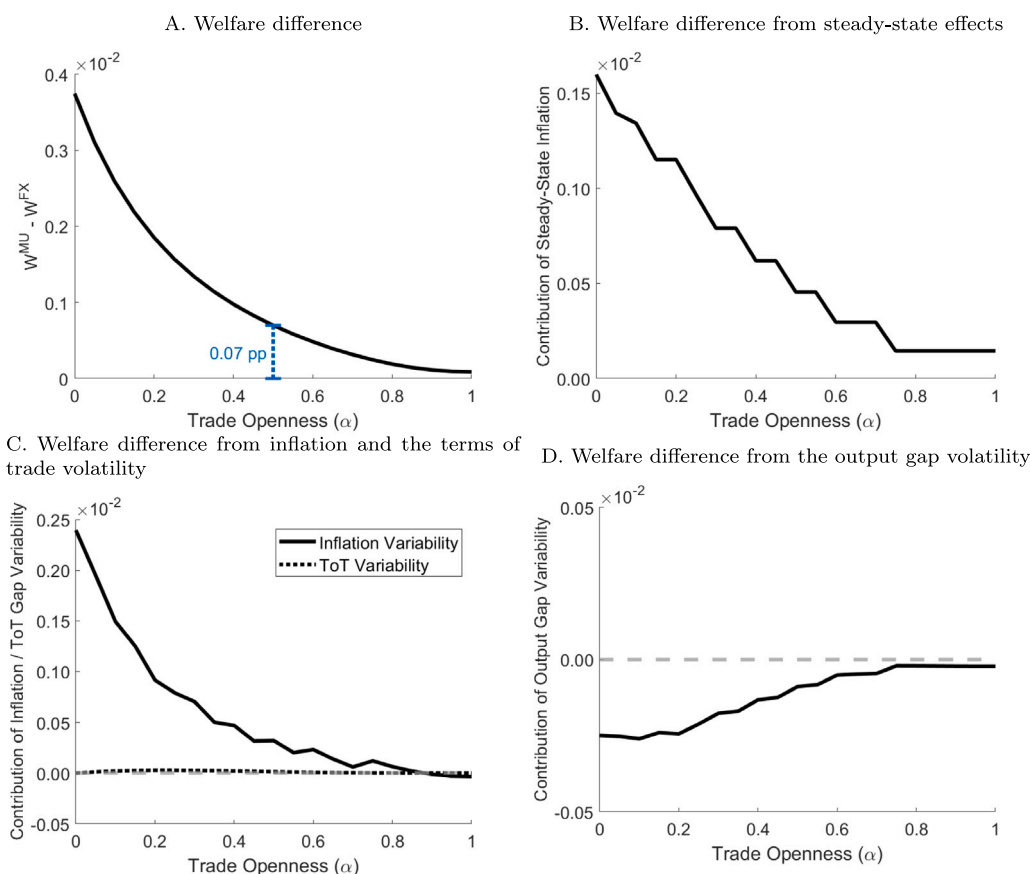


Fig. 6. Welfare difference (MU-FX) and its components.

Notes. The figure illustrates the welfare difference between a monetary union and a flexible exchange rate regime, as well as its components, for different degrees of trade openness α . Positive values indicate higher welfare under a monetary union. Panel A displays the overall welfare difference. Panel B shows the welfare difference arising from the steady-state effects. Panel C presents the welfare difference resulting from inflation and terms of trade volatility. Panel D depicts the welfare difference arising from output gap volatility.

Welfare implication. One of the most important questions in international macroeconomics is whether a flexible exchange rate regime is preferable to a monetary union. The international macroeconomic literature concludes that the desirability of flexible exchange rates depends on the presence of a ZLB (Cook and Devereux, 2016). That is, a flexible exchange rate regime is preferable to a monetary union when monetary policy is unconstrained, but the opposite is true when monetary policy is constrained. These conclusions are drawn under the assumption of a zero inflation target, where macroeconomic volatility is the sole determinant of welfare rank. When monetary policy is unconstrained, aggregate volatility is smaller under a flexible exchange rate regime than under a monetary union due to the favorable movement of the terms of trade, making the former desirable. However, when monetary policy is constrained, aggregate volatility under a flexible exchange rate regime increases as the terms of trade move perversely, making this regime undesirable. If central banks have the option to set the optimal inflation target, the welfare rank between the two exchange rate regimes does not depend solely on aggregate volatility but also on the cost of steady-state inflation.

We compare the welfare of the two exchange rate regimes, where each regime is set to its optimal inflation target. On the one hand, in the presence of the ZLB, raising the inflation target incurs a cost of steady-state price dispersion but can deliver reduced macroeconomic volatility. On the other hand, an attempt to curtail steady-state costs by choosing a low inflation target increases macroeconomic volatility. Thus, the welfare rank between the two exchange rate regimes is determined by this variance/steady-state trade-off, which does not exist under zero trend inflation. Panel A of Fig. 6 compares the welfare between the two exchange rate regimes at their respective optimal targets and reveals that a monetary union entails a higher level of welfare than a flexible exchange rate regime for all configurations of trade openness.

Panels B to D in Fig. 6 decompose the welfare difference between a monetary union and a flexible exchange rate regime into the components present in the loss function (17). Panel B shows that the welfare loss arising from steady-state price dispersion is smaller under a monetary union, reflecting the lower optimal inflation target in a monetary union than in a flexible exchange rate regime. Panel C shows that the welfare loss arising from inflation volatility is smaller under a monetary union, indicating that

inflation is more volatile in a flexible exchange rate regime than in a monetary union.¹⁵ As documented by Coibion et al. (2012), a higher inflation target induces more forward-looking behavior in firms when resetting their prices, resulting in larger inflation variations. As a result, a higher inflation target in a flexible exchange rate regime than in a monetary union leads to greater inflation volatility in the former. Panel D shows that the welfare loss arising from output gap variability is larger under a monetary union. Because of a higher optimal inflation target, a flexible exchange rate regime is subject to fewer ZLB episodes than a monetary union, contributing to smaller volatility of the output gaps in a flexible exchange rate regime. When observing Panels B, C, and D, we note that two factors contribute to the higher welfare of a monetary union: a lower steady-state cost and lower inflation volatility. Both these factors are attributable to a lower optimal inflation target in a monetary union. In our E.U. calibration ($\alpha = 0.5$), the formation of a monetary union increases per-period welfare by 0.07%, as measured by the percentage of efficient steady-state consumption, because it reduces the optimal inflation rate by 0.24%.

One might wonder whether our results on optimal inflation rates hold in a model with additional ingredients that the macroeconomic literature has found relevant in explaining realistic price dynamics. These ingredients include input–output linkages (Basu, 1995; Huang and Liu, 2001), financial frictions (Gilchrist et al., 2017, 2023), and endogenous price stickiness (L’Huillier and Schoenle, 2023). We discuss how these ingredients may affect our results in Appendix C and argue that the results presented in this section are likely to hold, at least qualitatively.

4. Optimal trend inflation when UIP breaks

The results in the preceding section emerged under the assumption that UIP holds in the data. Given the well-known empirical departures from the UIP (Fama, 1984), we now discuss how the results change when the UIP condition no longer holds. To break the UIP condition, we introduce an exogenous UIP wedge, similar to Itskhoki and Mukhin (2021). Due to the unclear guidance on empirically relevant UIP shock processes in the literature, we conduct two experiments: one with a small shock size that mildly breaks the UIP and the other with a large shock size that significantly breaks the UIP. In the case of a small UIP shock, the Fama coefficient computed from our calibrated model is not 1 but greater than 0, whereas a large shock results in a negative Fama coefficient. Experiments with different UIP shock sizes allow us to evaluate whether the degree of UIP violations matters for our open economy results.

To save space, we defer the model description and derivation of the loss function in the presence of UIP shocks to Sections 4 and 5 of the Online Appendix. Here, we provide the key equilibrium conditions affected by the introduction of UIP shocks and discuss how these shocks change aggregate demand, optimal inflation rates, and welfare. In the presence of UIP shocks, one can derive the modified UIP condition:

$$\mathbb{E}_t \Delta \hat{S}_{t+1} + \psi_t = \hat{R}_t - \hat{R}_t^*,$$

where the UIP shock follows:

$$\psi_t = \rho_\psi \psi_{t-1} + \epsilon_t^\psi \quad \text{with} \quad \epsilon_t^\psi \stackrel{iid}{\sim} \mathcal{N}(0, \sigma_\psi^2).$$

Solving the modified UIP condition forward, together with the definition of the terms of trade, yields the following expression for the terms of trade:

$$\hat{T}_t = -\mathbb{E}_t \sum_{j=0}^{\infty} \left[(\hat{R}_{t+j} - \hat{\pi}_{H,t+j+1}) - (\hat{R}_{t+j}^* - \hat{\pi}_{F,t+j+1}^*) \right] + \mathbb{E}_t \sum_{j=0}^{\infty} \psi_{t+j}. \quad (21)$$

$\psi_t = 0$ delivers a standard UIP condition, implying a Fama coefficient of 1. Under such a condition, Eq. (21) collapses to (18), and the terms of trade are determined solely by the difference in real interest rates between countries. A UIP shock also breaks the risk-sharing condition, yielding a modified risk-sharing condition:

$$\hat{Q}_t - \sigma (\hat{C}_t - \hat{C}_t^*) - z_{F,t} + z_{H,t} = \frac{\psi_t}{1 - \rho_\psi}, \quad (22)$$

where $\hat{Q}_t$ is the real exchange rate. Making use of this modified risk-sharing condition, Home net exports (20) change to:

$$n\hat{x}_t = \frac{\alpha}{2\sigma} [(2 - \alpha)(\sigma\eta - 1) + (1 - \sigma)]\hat{T}_t + \frac{\alpha}{2\sigma} (z_{F,t} - z_{H,t}) + \frac{\alpha}{2\sigma(1 - \rho_\psi)}\psi_t. \quad (23)$$

UIP shocks affect net exports in two ways. For illustration, consider a negative UIP shock, $\psi_t < 0$. First, it leads to an appreciation of the terms of trade, as described by Eq. (21). Then, the appreciation of the terms of trade changes net exports, where trade elasticity η plays a crucial role in determining the magnitude of net exports or their sign. Second, it acts as an exogenous friction in international financial markets in the sense that it counteracts the savings flow from Home to Foreign, which is initiated by negative Home demand shocks.

The quantitative importance of negative UIP shocks on Home aggregate demand depends greatly on the magnitude of the UIP shock. Let us consider negative UIP shocks with two different magnitudes: the UIP shock that is considerably smaller than the Home demand shock, $\frac{\psi_t}{(1 - \rho_\psi)} < z_{H,t}$, and the UIP shock that is large enough to completely counteract the negative demand shock,

¹⁵ We note that the welfare difference arising from the disparity in terms of trade gap volatility is negligible, as observed in Panel C.

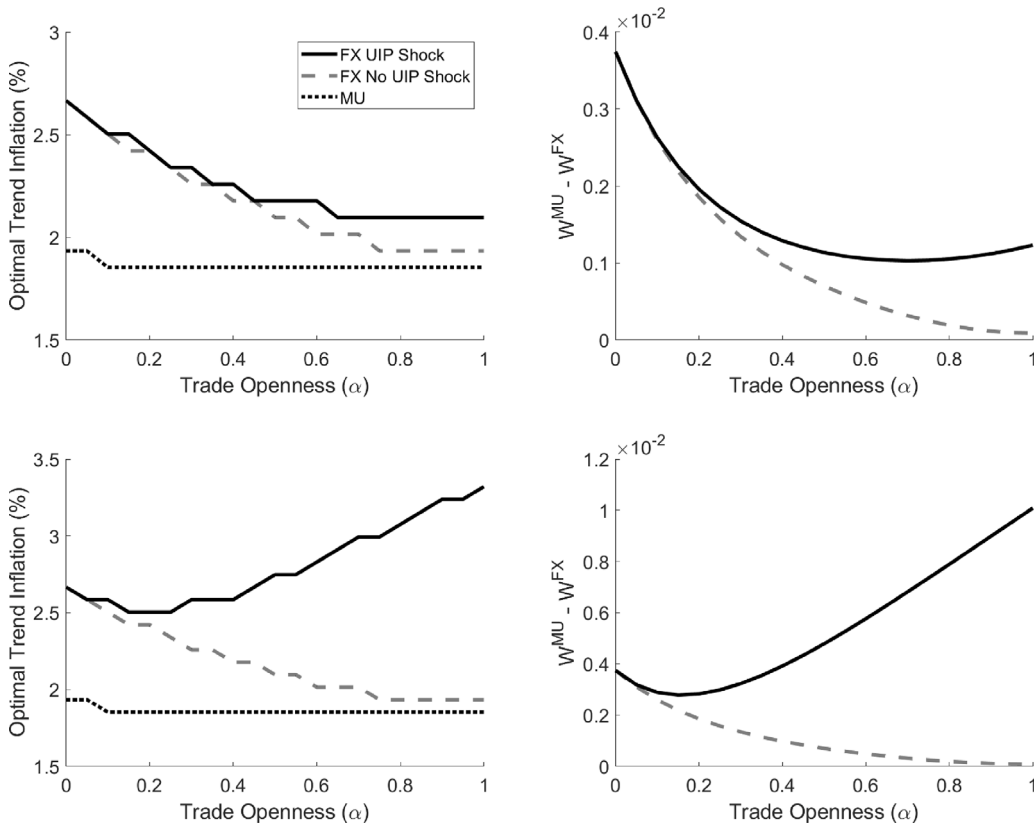


Fig. 7. Optimal inflation target, welfare, and UIP shocks.

Notes. The figure shows the effect of UIP shocks on the optimal inflation rates and the relative welfare of a monetary union. Row 1 corresponds to the small UIP shock scenario, and row 2 corresponds to the large UIP shock scenario.

$\frac{\psi_t}{(1-\rho_\psi)} = z_{H,t}$. In the small-shock case, net exports are still largely driven by the negative demand shock, so Home is still a net saver, as in the frictionless economy described in Section 2. Thus, with an increase in trade openness α , the flow of savings from Home to Foreign (i.e., net exports) increases, mitigating the deflationary pressure at the ZLB. However, because the small negative UIP shock partially offsets the savings flow from Home to Foreign, the deflationary pressure is stronger than in a scenario where this shock is absent for a given α . Notice that the direct offsetting force of the UIP shocks increases with trade openness α , as captured by the coefficient $\frac{\alpha}{2\sigma(1-\rho_\psi)}$ in Eq. (23).

In the large-shock case, the direct savings flows induced by negative demand shocks are completely offset. In this case, net exports are driven purely by the terms of trade, with the magnitude or sign depending on trade openness α and trade elasticity η . If $\eta > 1$, given $\sigma = 1$, the fall in net exports in response to terms of trade appreciation increases as trade openness α increases, as captured by coefficient $\frac{\alpha}{2\sigma}[(2-\alpha)(\sigma\eta-1) + (1-\sigma)]$, which increases with α . In other words, savings flow from Home to Foreign decreases as trade openness increases, generating more severe deflationary pressure at the ZLB. In this case, one would expect more contracted aggregate demand as trade openness increases. If $\eta = 1$, net exports are zero, and thus, the aggregate demand would be identical to that in a closed economy, so aggregate demand is unaffected by trade openness. If $\eta < 1$, the sign of the net exports in response to terms of trade appreciation is positive, suggesting that Home savings flow to Foreign. In this case, Home net exports exhibit an increasing pattern with respect to trade openness. This implies that the savings flow from Home to Foreign increases as trade openness increases, easing the deflationary pressure at the ZLB. Thus, one would expect less contracted aggregate demand as trade openness increases.¹⁶

Results. Following Itskhoki and Mukhin (2021), the persistence of the UIP shock ρ_ψ is set equal to that of the fundamental shock, which is a demand shock in our model. In the experiment that assumes small UIP shocks, we set their standard deviation, σ_ψ , to 0.003, resulting in a Fama coefficient of 0.71. Regarding the experiment with large UIP shocks, we set $\sigma_\psi = 0.009$, which delivers a Fama coefficient of -0.12 . The parameterization of the remaining shock processes remains unchanged from the baseline values.

¹⁶ In Appendix A, we analyze the sensitivity of the optimal inflation rates to trade elasticity when UIP breaks. Trade elasticity does not matter qualitatively when departure from UIP is mild but matters substantially when departure is significant.

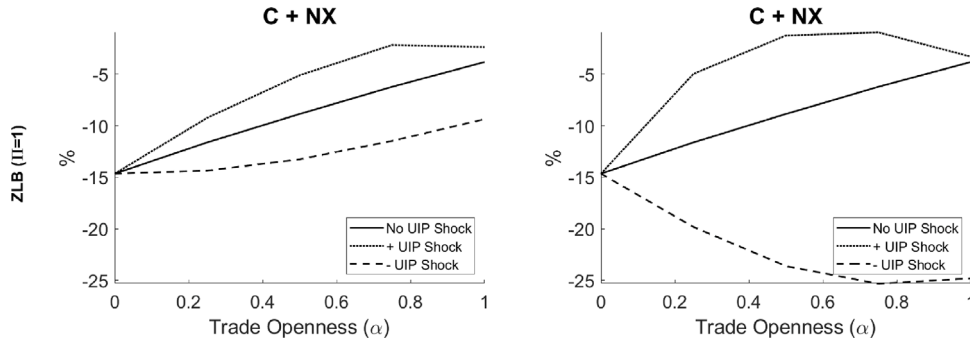


Fig. 8. Aggregate demand under UIP shocks.

Notes. The figure shows the effect of UIP shocks on Home aggregate demand under zero trend inflation. The left panel corresponds to the small UIP shock scenario, and the right panel shows the large UIP shock scenario.

Row 1 in Fig. 7 illustrates the optimal trend inflation and relative welfare of a monetary union in the case of small UIP shocks compared to those without such shocks. In the figure, ‘FX UIP Shock’ corresponds to the case with small UIP shocks under flexible exchange rates, implying a mild departure from UIP. ‘FX No UIP Shock’ corresponds to the case without UIP shocks under flexible exchange rates, namely a scenario in which UIP holds. As identified by comparing $\alpha > 0$ and $\alpha = 0$, the optimal inflation rate in an open economy is still lower than in a closed economy when departures from UIP are mild. Moreover, for each level of trade openness α , the optimal trend inflation in a flexible exchange rate regime becomes higher when the UIP breaks than when it does not, thus enhancing the desirability of a monetary union.

The reason for the higher optimal inflation target under flexible exchange rates when UIP breaks originates from the disproportionate impact of negative UIP shocks on welfare. The left panel of Fig. 8 compares the Home aggregate demand response, which is the sum of consumption and net exports, on impact at the ZLB in scenarios where UIP holds and does not. When UIP holds (solid lines), the underlying shocks are negative demand shocks that cause the Home ZLB to bind for 10 periods when trade openness α is zero and trend inflation is zero. When UIP does not hold, there can be either positive or negative UIP shocks, in addition to negative demand shocks, owing to the uncorrelated shock structure between UIP and demand shocks. The figure shows that except in a closed economy setting, positive UIP shocks (dotted lines) mitigate the contraction in aggregate demand caused by negative demand shocks, whereas negative UIP shocks (dashed lines) aggravate the contraction. Mitigated aggregate demand contraction due to positive UIP shocks alleviates deflationary pressure, whereas aggravated contraction stemming from negative UIP shocks increases deflationary pressure. Due to the convexity of the welfare loss function with respect to inflation variability, the welfare loss associated with a more severe aggregate demand contraction exceeds the welfare gain from the mitigated contraction. Therefore, a departure from UIP is welfare-costly at the ZLB, and, hence, a higher increase in the optimal inflation target is necessary to avoid such a costly situation. Because the optimal inflation target diverges further from that in a monetary union when UIP breaks, a flexible exchange rate regime becomes more welfare detrimental to a monetary union. Under our calibrated trade openness ($\alpha = 0.5$), forming a monetary union results in a decrease in the optimal inflation rate and an increase in per-period welfare by 0.32% and 0.11%, respectively.

What is the mechanism behind the more contracted aggregate demand associated with negative UIP shocks? As can be inferred from Eq. (23), a negative UIP shock depresses the aggregate demand at the ZLB in two ways. First, according to (21), it exacerbates the perverse movement of the terms of trade resulting from negative Home demand shocks. The exacerbated terms of trade intensify the expenditure switching away from Home goods, thus suppressing Home net exports. The effect of a negative UIP shock on net exports which works through the terms of trade is captured by $\frac{\alpha}{2\sigma}[(2-\alpha)(\sigma\eta-1)+(1-\sigma)]\hat{T}_t$ in Eq. (23). Second, a negative UIP shock directly counters the savings flow from Home to Foreign (i.e., Home net exports). The direct effect of negative UIP shocks on net exports is captured by $\frac{\alpha}{2\sigma(1-\rho_\psi)}\psi_t$ in Eq. (23). Because of the offsetting role of a negative UIP shock on savings, the efficacy of trade openness in mitigating deflationary pressure diminishes, and, thus, the Home real interest rate at the ZLB increases more than when UIP holds. Consequently, Home consumption decreases more than it does in the absence of negative UIP shocks. Despite the more depressed aggregate demand in the presence of small negative UIP shocks, the optimal inflation rate still exhibits a decreasing pattern with trade openness α . This is primarily because net exports are driven predominantly by the direct effect of demand shocks, which have a larger magnitude than UIP shocks. As the offsetting effect of small negative UIP shocks increases with trade openness α , the difference in aggregate demand between the cases with and without negative UIP shocks widens, explaining the increasing gap in optimal inflation rates as trade openness increases.

Row 2 of Fig. 7 displays the optimal inflation and relative welfare of a monetary union in the case of large UIP shocks compared with those without such shocks. In comparison to the small shock experiment in row 1, the optimal trend inflation under a flexible exchange rate is higher for all values of $\alpha > 0$. Importantly, unlike in the small shock experiment, the optimal inflation rate exhibits an increasing pattern with trade openness, so an open economy with a sufficiently large degree of trade openness delivers a higher optimal inflation rate than a closed economy. The relative welfare of a monetary union exhibits a similar pattern, reflecting the cost of increasing inflation targets. The increasing optimal inflation rate with trade openness follows from the right panel of Fig. 8, which illustrates the decreasing aggregate demand associated with a large negative UIP shock at the ZLB. Again, due to the convexity of

the welfare loss function, the welfare loss from aggregate demand contraction dominates the welfare gain from aggregate demand stabilization, and this dominance becomes more pronounced as trade openness increases. Therefore, a higher optimal inflation rate is warranted in an open economy than in a closed economy, and the relative welfare of a monetary union increases more than when the size of the UIP shocks is small. Given our baseline α of 0.5, forming a monetary union reduces the optimal inflation rate by 0.9% and increases per-period welfare by 0.48%, when UIP deviations are significant.

Why does the aggregate demand decrease with trade openness when UIP shocks are large and negative? As an intuitive explanation, suppose that the magnitude of a negative UIP shock is sufficiently large to offset the direct effect of negative demand shocks. In this scenario, $\frac{\alpha}{2\sigma}(z_{F,t} - z_{H,t}) + \frac{\alpha}{2\sigma(1-\rho_\psi)}\psi_t$ in Eq. (23) becomes zero, so Home net exports are determined solely by the terms of trade. With large UIP shocks, the terms of trade appreciate more than when these shocks are small, as shown in Eq. (21). The more pronounced appreciation of the terms of trade affects Home net exports, with the effect depending on the coefficient of the terms of trade in Eq. (23), $\frac{\alpha}{2\sigma}[(2-\alpha)(\sigma\eta-1) + (1-\sigma)]$. If $\eta > 1$ and $\sigma = 1$, which is the case in our baseline calibration, the sign of the coefficient is positive and increases with trade openness α . In this case, Home net exports fall and exhibit a decreasing pattern with respect to trade openness. Decreasing Home net exports implies that Home's net borrowing position increases with trade openness. In other words, it becomes more difficult for savings to flow from Home to Foreign as trade openness increases, exacerbating the deflationary pressure at the ZLB. Consequently, Home real interest rates increase with trade openness, generating a decreasing pattern in Home consumption as trade openness increases.

5. Conclusion

This paper derives the optimal inflation target in an open economy using a two-country model calibrated to the Euro area. When UIP holds, the optimal inflation target for an open economy is smaller than that for a closed economy, regardless of the exchange rate regime. As trade openness increases, savings in a country hit by an adverse demand shock flow more to the partner country in the form of net exports, resulting in less contracted output than in a closed economy. Consequently, the deflationary expectations are less severe, causing a smaller increase in real interest rates at the ZLB. Thus, trade openness eases the downward pressure on aggregate demand and reduces the cost of ZLB. In addition, the optimal inflation target is higher under a flexible exchange rate regime than under a monetary union because the aggregate demand contraction at the ZLB is less severe in a monetary union. This less severe aggregate demand contraction in a monetary union originates from the sluggish dynamics of the terms of trade, which make net exports higher than in a flexible exchange rate regime, and from the price-level targeting-like effect, which alleviates the decline in consumption. A lower optimal inflation target for a monetary union suggests that it delivers higher welfare than a flexible exchange rate regime.

When UIP does not hold, the optimal inflation target in an open economy becomes higher than when UIP holds and can be higher than in a closed economy if the deviation from UIP is significant. Moreover, as departures from UIP become more pronounced, the optimal inflation target in a flexible exchange rate regime increases more than in a monetary union, thereby enhancing the desirability of a monetary union. The increasing optimal inflation target with departure from UIP results from the UIP shock, which exerts further downward pressure on the aggregate demand at the ZLB.

Data availability

No data was used for the research described in the article.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.jmoneco.2024.103675>.

References

- Adam, K., Weber, H., 2019. Optimal trend inflation. *Amer. Econ. Rev.* 109, 702–737.
- Basu, S., 1995. Intermediate goods and business cycles: Implications for productivity and welfare. *Am. Econ. Rev.* 85 (3), 512–531.
- Basu, S., Fernald, J.G., 1997. Returns to scale in U.S. production: Estimates and implications. *Q. J. Econ.* 105 (2), 249–283.
- Bayoumi, T., Eichengreen, B., 1993. Shocking aspects of European monetary integration. In: Torres, F., Giavazzi, F. (Eds.), *Adjustment and Growth in the European Monetary Union*. Cambridge University Press, pp. 193–235.
- Bhattarai, S., Eggertsson, G.B., Schoenle, R., 2018. Is increased price flexibility stabilizing? *Redux. J. Monetary Econ.* 100, 66–82.
- Bils, M., Klenow, P.J., 2004. Some evidence on the importance of sticky prices. *J. Polit. Econ.* 112 (5), 947–985.
- Blanchard, O., 2022. It is time to revisit the 2% inflation target. *Financial Times* November 29.
- Blanco, A., 2021. Optimal inflation target in an economy with menu costs and a zero lower bound. *Am. Econ. J.: Macroecon.* 13 (3), 108–141.
- Broda, C., Weinstein, D.E., 2006. Globalization and the gains from variety. *Q. J. Econ.* 121 (2), 541–585.
- Campos, N.F., Macchiarelli, C., 2016. Core and periphery in the European monetary union: Bayoumi and eichengreen 25 years later. *Econom. Lett.* 147, 127–130.
- Carreras, M., Coibion, O., Gorodnichenko, Y., Wieland, J., 2016. Infrequent but long-lived zero lower bound episodes and the optimal rate of inflation. *Annu. Rev. Econ.* 8 (1), 497–520.
- Cho, D., Kim, K.H., Kim, S.J., 2023. The paradox of price flexibility in an open economy. *Rev. Econ. Dyn.* 51, 370–392.
- Coibion, O., Gorodnichenko, Y., Wieland, J., 2012. The Optimal Inflation Rate in New Keynesian Models: Should Central Banks Raise Their Inflation Targets in Light of the Zero Lower Bound?. *Rev. Econ. Stud.* 79 (4), 1371–1406.
- Cook, D., Devereux, M.B., 2013. Sharing the burden: Monetary and fiscal responses to a world liquidity trap. *Am. Econ. J.: Macroecon.* 5 (3), 190–228.
- Cook, D., Devereux, M.B., 2016. Exchange rate flexibility under the zero lower bound. *J. Int. Econ.* 101, 52–69.
- Corsetti, G., Dedola, L., Leduc, S., 2011. Optimal monetary policy in open economies. *Handb. Monet. Econ.* 3, 861–933.

- Corsetti, G., Dedola, L., Leduc, S., 2023. Exchange rate misalignment and external imbalances: What is the optimal monetary policy response?. *J. Int. Econ.* 144, 103771.
- Eggertsson, G.B., Krugman, P., 2012. Debt, deleveraging, and the liquidity trap: A Fisher-minsky-koo approach. *Q. J. Econ.* 127 (3), 1469–1513.
- Eggertsson, G.B., Woodford, M., 2003. The zero bound on interest rates and optimal monetary policy. *Brook. Pap. Econ. Act.* 2003 (1), 139–211.
- Engel, C., 2011. Currency misalignments and optimal monetary policy: A reexamination. *Amer. Econ. Rev.* 101 (6), 2796–2822.
- Fama, E.F., 1984. Forward and spot exchange rates. *J. Monetary Econ.* 14 (3), 319–338.
- Fujiwara, I., Nakajima, T., Sudo, N., Teranishi, Y., 2013. Global liquidity trap. *J. Monetary Econ.* 60 (8), 936–949.
- Gilchrist, S., Schoenle, R., Sim, J., Zakrajsek, E., 2017. Inflation dynamics during the financial crisis. *Amer. Econ. Rev.* 107 (3), 785–823.
- Gilchrist, S., Schoenle, R., Sim, J., Zakrajsek, E., 2023. Financial heterogeneity and monetary union. *J. Monetary Econ.* 139, 21–40.
- Groll, D., Monacelli, T., 2020. The inherent benefit of monetary unions. *J. Monetary Econ.* 111, 63–79.
- Hohberger, S., Priftis, R., Vogel, L., 2019. The macroeconomic effects of quantitative easing in the euro area: Evidence from an estimated DSGE model. *J. Econom. Dynam. Control* 108, 103756.
- Holden, T.D., 2016. Computation of Solutions to Dynamic Models with Occasionally Binding Constraints. (130143), ZBW - Leibniz Information Centre for Economics.
- Holden, T.D., 2023. Existence and Uniqueness of Solutions to Dynamic Models with Occasionally Binding Constraints. *Rev. Econ. Stat.* 105 (6), 1481–1499.
- Huang, K.X.D., Liu, Z., 2001. Production Chains and General Equilibrium Aggregate Dynamics. *J. Monetary Econ.* 48 (2), 437–462.
- Itskhoki, O., Mukhin, D., 2021. Exchange Rate Disconnect in General Equilibrium. *J. Polit. Econ.* 129 (8), 2183–2232.
- Kim, J., Ruge-Murcia, F.J., 2009. How much inflation is necessary to grease the wheels?. *J. Monetary Econ.* 56 (3), 365–377.
- Krugman, P., 1998. It's baaack: Japan's slump and the return of the liquidity trap. *Brook. Pap. Econ. Act.* 29 (2), 137–206.
- L'Huillier, J.-P., Schoenle, R., 2023. Raising the inflation target: What are the effective gains in policy room?. Working Paper.
- Lubik, T., Schorfheide, F., 2005. A Bayesian look at new open economy macroeconomics. *NBER Macroecon. Annu.* 20, 313–382.
- Mineyama, T., 2022. Revisiting the optimal inflation rate with downward nominal wage rigidity: The role of heterogeneity. *J. Econom. Dynam. Control* 139, 104350.
- Nakamura, E., Steinsson, J., 2008. Five facts about prices: A reevaluation of menu cost models. *Q. J. Econ.* 123 (4), 1415–1464.
- Pappa, E., 2004. Do the ECB and the fed really need to cooperate? Optimal monetary policy in a two-country world. *J. Monetary Econ.* 51 (4), 753–779.
- Simonovska, I., Waugh, M., 2014. Elasticity of trade: Estimates and evidence. *J. Int. Econ.* 92 (1), 34–50.
- Smets, F., Wouters, R., 2003. An estimated dynamic stochastic general equilibrium model of the euro area. *J. Eur. Econom. Assoc.* 1 (5), 1123–1175.
- Zhao, H., 2022. On the impacts of trend inflation in an open economy. *J. Int. Econ.* 138, 103648.